LYLE OKOTH

MACHINE LEARNING RESEARCHER - SPEECH SYNTHESIS AND AUDIO AI

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PROFESSIONAL SUMMARY

Machine Learning Researcher with **6+ years** building speech systems for Kenyan and other under-resourced languages, from raw field recordings to published models. Specialises in **diffusion and flow-matching TTS**, fine-tuned ASR and speaker diarization, plus the evaluation methodology that makes a result defensible.

Trained a **0.2s RTF synthesizer covering 8 Kenyan languages** on one consumer GPU, cut Swahili/Sheng **CER from 21% to 9%**, and released **4 models and 3 datasets with 12,000+ Hugging Face downloads**.

TECHNICAL SKILLS

- **Speech and audio:** PyTorch, torchaudio, Hugging Face Transformers, Whisper, VITS, F5-TTS, pyannote.audio, librosa, soundfile, forced alignment, voice cloning
- **Model training:** Diffusion and flow-matching TTS, LoRA / PEFT, int8 quantization, DeepSpeed, Accelerate, Datasets, CUDA, multi-GPU fine-tuning, preference tuning and guardrail pipelines
- **Evaluation and research ops:** WER / CER / MOS, DER, real-time factor, held-out test sets, ablation studies, Weights & Biases, MLflow, model cards, reproducible training recipes
- **Speech data:** Corpus building, consented field recording, noise and silence filtering, loudness normalisation, Label Studio, reviewer agreement, FFmpeg, DuckDB, Parquet, Apache Airflow, S3
- **Delivery:** FastAPI, ONNX Runtime, Docker, GitHub Actions, AWS, Python, SQL, JavaScript

WORK EXPERIENCE

Machine Learning Researcher | SautiFlow Labs — 2026 - Present

Diffusion and flow-matching speech synthesis for languages with almost no public data.

- Trained a flow-matching TTS model covering **8 Kenyan languages** at **0.2s real-time factor** on a single consumer GPU, bringing multilingual synthesis inside interactive latency budgets.
- Cut **character error rate from 21% to 9%** on in-house Swahili and Sheng evaluation sets through fine-tuning, corpus curation and accent-aware text normalisation.
- Built a **120-hour** transcribed and force-aligned speech corpus from raw field recordings - the training base for the lab's synthesis and recognition models.
- Reduced **VRAM by 45%** with int8 quantization at a **0.4% accuracy cost**, cutting inference hardware per deployed voice.
- Published **4 models and 3 datasets on Hugging Face (12,000+ downloads)** with model cards, training recipes and reproducible evaluation baselines.
- Ran ablations tracked in Weights & Biases, reporting **WER / CER / MOS** per release so model choices hold up against a held-out test set.

AI Product Developer | Freelance — 2026 - Present

End-to-end ownership on a small number of products: research, flow design, model, launch, pricing, operation.

- Built **Voxa** alone from brief to live product: speech-skit generation, account system, **3 billing paths** and a studio interface for creators.
- Made infrastructure portable by design - storage swappable across local disk, **R2, S3 and Azure** with **no application code changes**.

Machine Learning Engineer | SautiFlow Labs **2022 - 2026**

Owned the path from checkpoint to service: pipelines, inference APIs, evaluation harnesses, cost control.

- Shipped the SemaScribe speaker-diarization pipeline, processing **50,000 hours of meeting audio at 12% diarization error rate**.
- Served speech inference on FastAPI + Celery + Redis at **8,000 requests/day at 180ms p95**, using ONNX Runtime, dynamic batching and GPU worker pools.
- Cut **training spend by 40%** with mixed precision and preemptible GPU capacity at **equal model quality**, verified on frozen evaluation sets.
- Automated ingestion of **200 hours of audio per week** with **98% label agreement across 3 reviewers** via ASR pre-labelling and reviewer rounds.

Software and Data Engineer | SautiFlow Labs **2022 - 2025**

Two sides of one job: move data reliably, and ship a product people can actually use.

- Built SnapSavanna, a dataset and labeling platform hosting **18 African-language datasets and 40,000 labeled samples** with content-hash versioning.
- Designed domain-driven FastAPI services split into **5 bounded contexts** with ports and adapters at the edges, keeping model code out of the product path.

SELECTED PROJECTS

Vocalis - Voice Design and Speech Studio

Flow-matching TTS studio: brand voice design, cloning from short consented samples, 24/48 kHz masters, consent checks and audible watermarking, GPU inference queue with streamed output.

Tech: PyTorch, torchaudio, VITS, Whisper, ONNX Runtime, FastAPI, Redis, Celery, React, Kubernetes

EchoiID - Speaker Diarization and ASR

Who-spoke-when plus per-speaker transcription on stored files or live audio: voiceprint embeddings, clustering with overlap handling, word timestamps, SRT/VTT/JSON export.

Tech: PyTorch, pyannote.audio, Whisper, FastAPI, WebSockets, Redis, Celery, PostgreSQL, Docker, AWS

EDUCATION

BSc Computer Security and Forensics | Jaramogi Oginga Odinga University of Science and Technology (JOOUST) **[20XX - 20XX]**

ML, signal processing and security coursework; final-year project on audio watermarking for corpus provenance.

Mechatronics Engineering | Jomo Kenyatta University of Agriculture and Technology (JKUAT) **[20XX - 20XX]**

Fourier analysis, sampling theory, filter design and real-time embedded constraints - the grounding behind latency work.

OPEN SOURCE, WRITING AND COMMUNITY

- **Open ML releases:** 4 models and 3 datasets on Hugging Face, **12,000+ downloads**; model cards and training recipes per release.
- **Writing and tooling:** speech AI articles on Medium (@lyle-okoth); Python libraries for the YouTube and Google Drive APIs on PyPI.
- **Community:** OMDENA volunteer (2024); co-founder, SautiFlow Labs (2022).